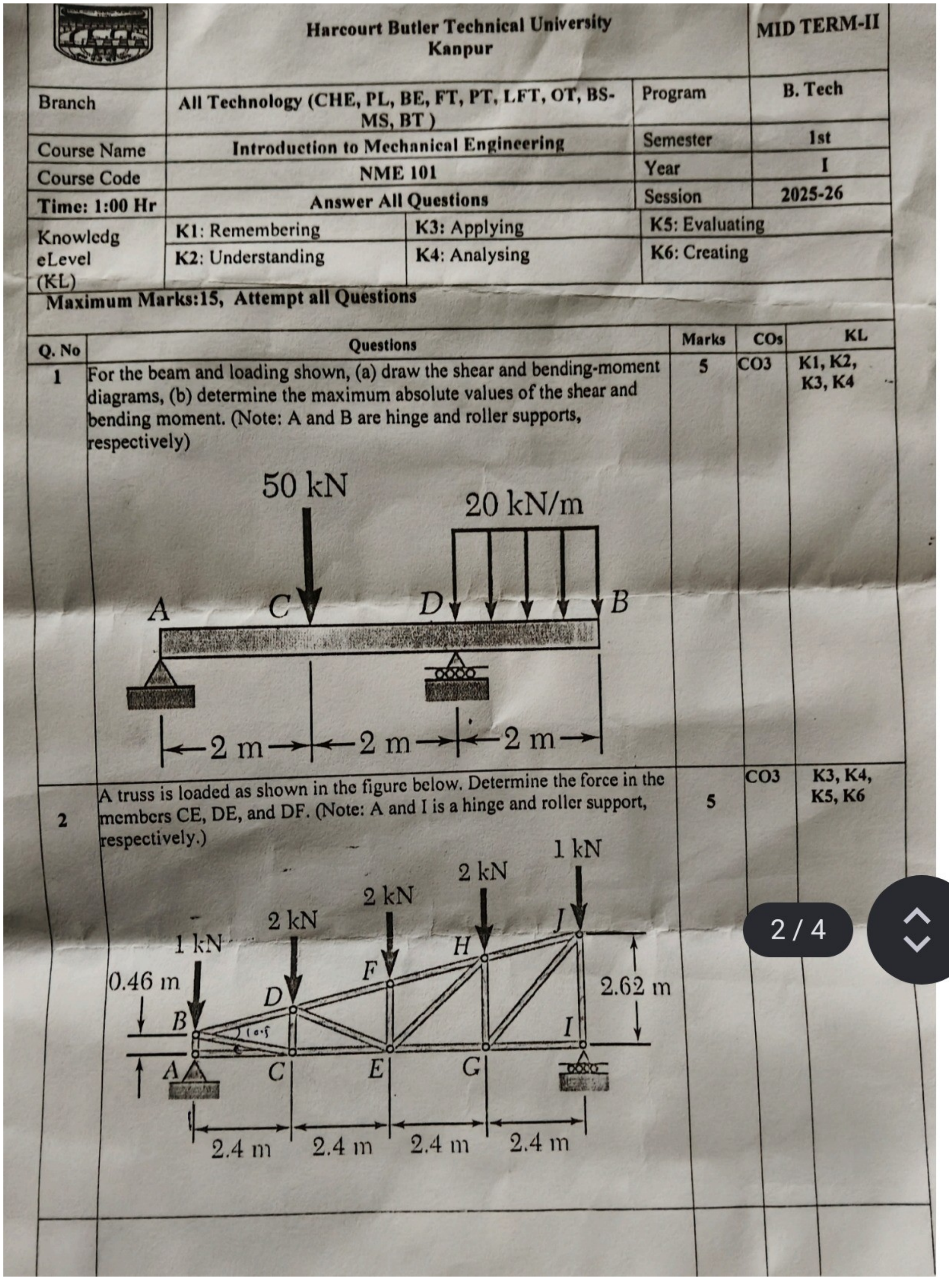
UNIT 3: Beams and Trusses

Syllabus topics: *Shear force & bending moment; differential equations of equilibrium; SFD & BMD for statically determinate beams (simply supported & cantilever); trusses — method of joints & method of sections.*

Question 1



	Harcourt Butler Technical University Kanpur		MID TERM-II
Branch	All Technology (CHE, PL, BE, FT, PT, LFT, OT, BS-MS, BT)	Program	B. Tech
Course Name	Introduction to Mechanical Engineering	Semester	1st
Course Code	NME 101	Year	I
Time: 1:00 Hr	Answer All Questions	Session	2025-26

Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating
	K2: Understanding	K4: Analysing	K6: Creating

Maximum Marks:15, Attempt all Questions

Q. No	Questions	Marks	COs	KL
1	For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment. (Note: A and B are hinge and roller supports, respectively)	5	CO3	K1, K2, K3, K4
2	A truss is loaded as shown in the figure below. Determine the force in the members CE, DE, and DF. (Note: A and I is a hinge and roller support, respectively.)	5	CO3	K3, K4, K5, K6

MID TERM-II | NME 101 | Q1 Beam SFD/BMD (hinge & roller), Q2 Truss – force in members CE, DE, DF

Question 2

Level (KL)	K2:Understanding	K4:Analysing	K6:Creating

Q. No	Questions	Marks	COs	KL
1a	A beam 10 m long and simply supported at each end, has a uniformly distributed load of 1000 N/m extending from the left end upto the centre of the beam. There is also an anti-clockwise couple of 15 kNm at a distance of 2.5 m from the right end. Draw the S.F. and B.M. diagrams.	5	CO3	K3
1b	Determine the forces in the truss shown in Figure 1 which carries a horizontal load of 16 kN and a vertical load of 24 kN.	5	CO3	K4, K5
2b	Determine the C.G. of the uniform plane lamina shown in Fig. 2. All dimensions are in cm.	5	CO4	K3

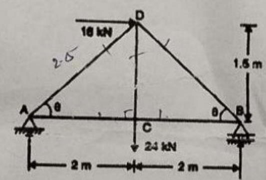
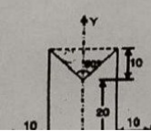


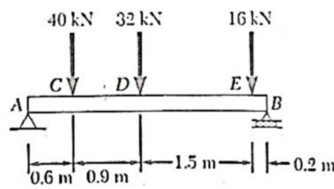
Figure 1.



MID TERM-II | NME 102 | 2025-26 | Q1(a) Beam with UDL & couple – SFD/BMD, Q1(b) Truss with horizontal & vertical load

Question 3

3a	For the beam and loading shown, (*a*) draw the shear and bending-moment diagrams, (*b*) determine the maximum absolute values of the shear and bending moment.	5	CO3	K2, K3, K4, K5



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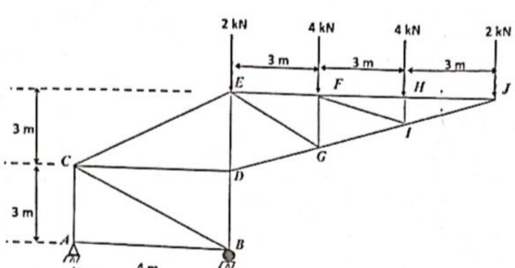
NME 101 | 2025-26 | Q3(a) Beam SFD/BMD with point loads 40kN, 32kN, 16kN

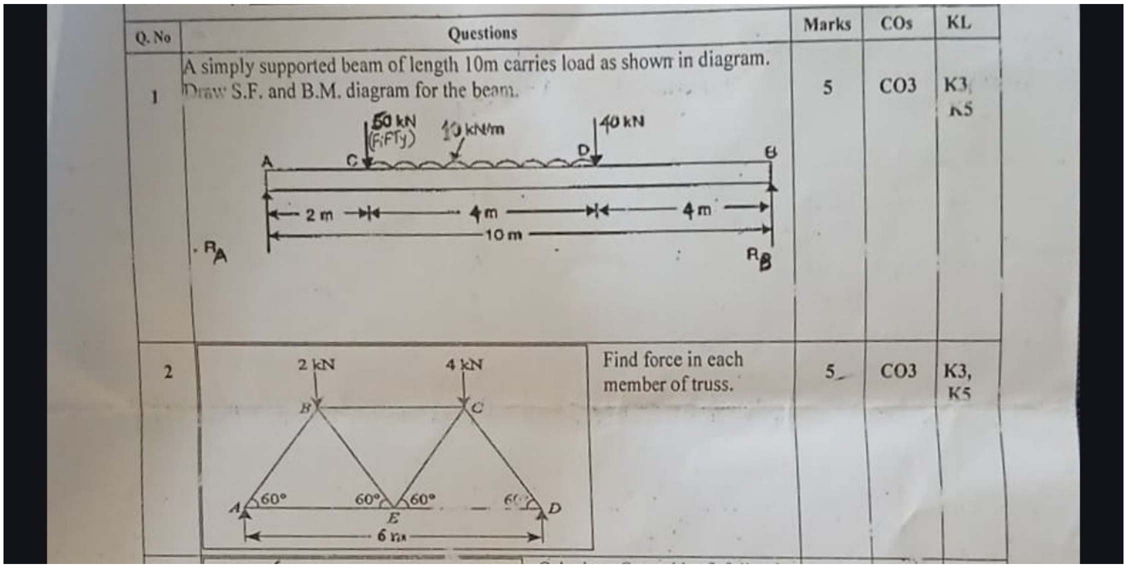
Question 4

3b	Determine the force in members CE, CD and CB of the truss shown in the figure below.	5	CO3	K2, K3, K4, K5
4a	State and prove the parallel-axis theorem			



NME 101 | 2025-26 | Q3(b) Truss – force in members CE, CD, CB

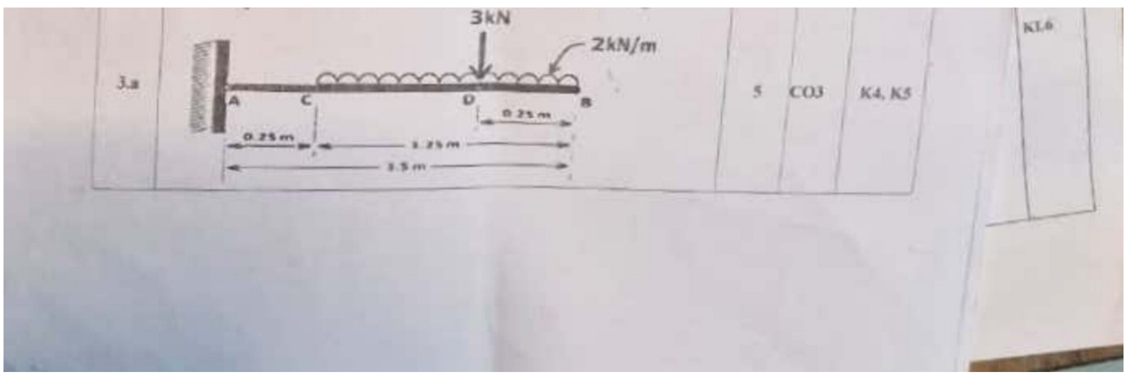
Question 5



Q. No	Questions	Marks	COs	KL
1	A simply supported beam of length 10m carries load as shown in diagram. Draw S.F. and B.M. diagram for the beam.	5	CO3	K3, K5
2	Find force in each member of truss.	5	CO3	K3, K5

2nd MID SEM | NME-101 | 2024-25 | Q1 Simply supported beam SFD/BMD, Q2 Truss – force in each member

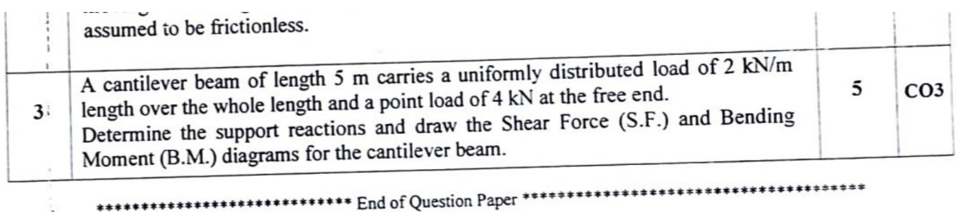
Question 6



3.a		5	CO3	K4, K5	KL6

END SEM | NME-101 | Q3(a) Cantilever beam SFD/BMD with point load & UDL

Question 7



	assumed to be frictionless.		
3	A cantilever beam of length 5 m carries a uniformly distributed load of 2 kN/m length over the whole length and a point load of 4 kN at the free end. Determine the support reactions and draw the Shear Force (S.F.) and Bending Moment (B.M.) diagrams for the cantilever beam.	5	CO3

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1st MID SEM | NME 101 | Q3 Cantilever beam with UDL & point load at free end

Question 8

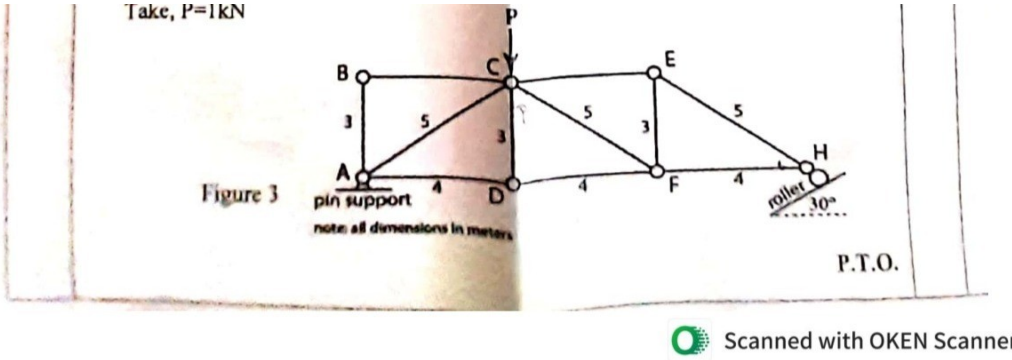
Take, P=1kN

Figure 3

P.T.O.

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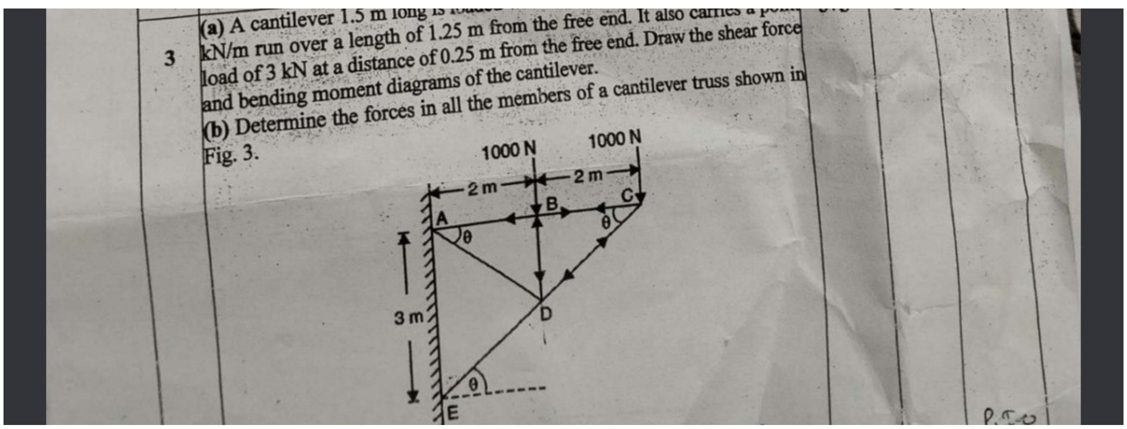
END SEM | NME-102 | 2022-23 | Q3(a) Truss – support reactions & forces in AD, CF, CE

Question 9

3(b)	Derive the expression for differential equations of equilibrium relating rate of loading, shear force and bending moment. Discuss the geometrical interpretation of equations with example.	5	CO3

END SEM | NME-102 | 2022-23 | Q3(b) Differential equations of equilibrium (loading, SF, BM)

Question 10


3 | **(a)** A cantilever 1.5 m long is loaded ... kN/m run over a length of 1.25 m from the free end. It also carries a point load of 3 kN at a distance of 0.25 m from the free end. Draw the shear force and bending moment diagrams of the cantilever.

(b) Determine the forces in all the members of a cantilever truss shown in Fig. 3.

P.T.O

End Semester | Q3(a) Cantilever beam SFD/BMD, Q3(b) Forces in cantilever truss members